

LPS002 - Storm Report

October 8, 2026

1 Observations of The Regulus Project

Focus: Historical Audit: 2026-10-02 12:00:00 to 2026-10-07 0:59:59

1.1 Recorded Observation Highlights

Observations from 2026-10-02 12:00:00 (UTC) to 2026-10-07 0:59:59 (UTC)

Min Pressure 1013.21 mb

Max Pressure 1019.87 mb

Max Wind Speed 24.18 mph

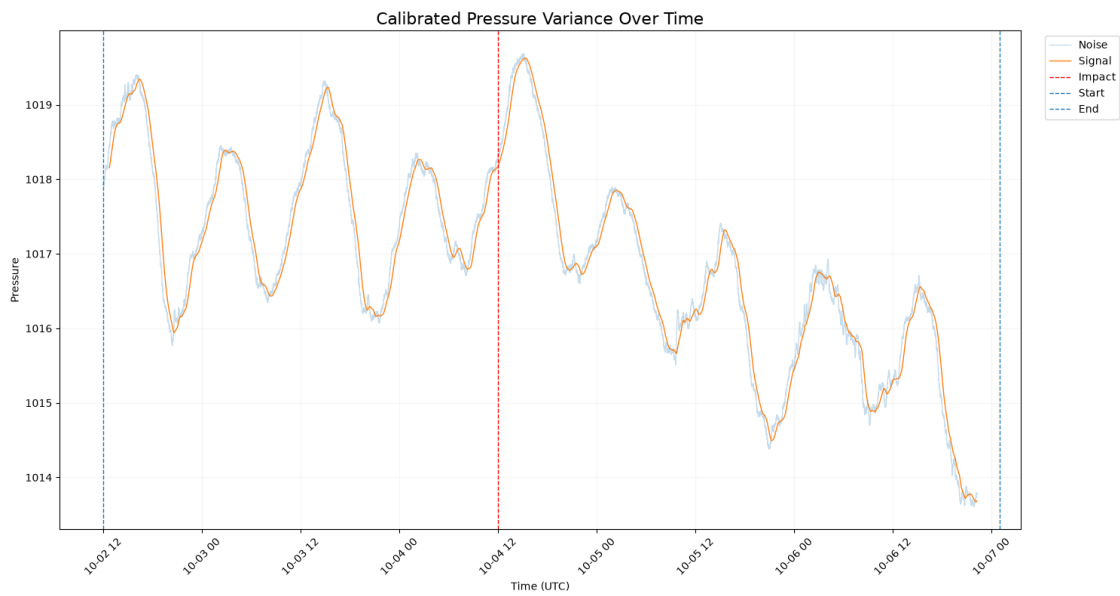
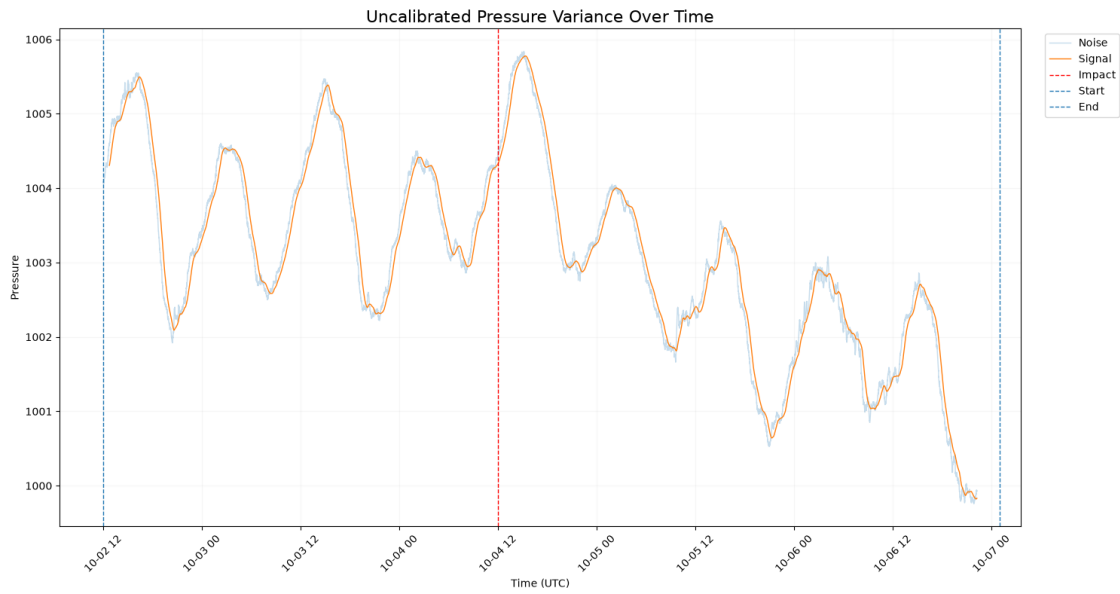
Total Rainfall 3.78 in (96.02 mm)

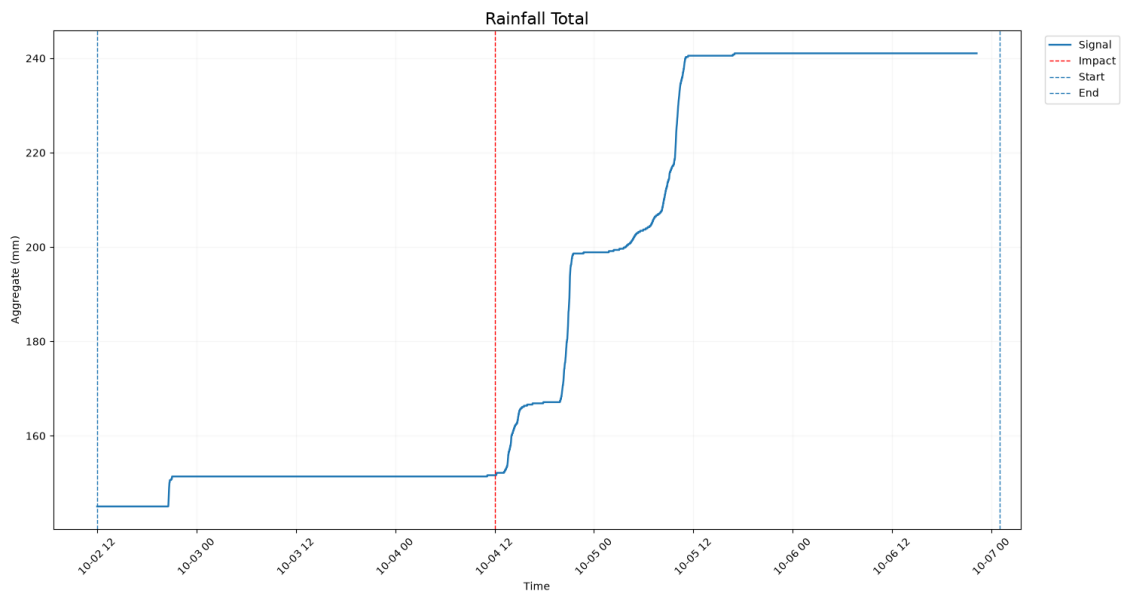
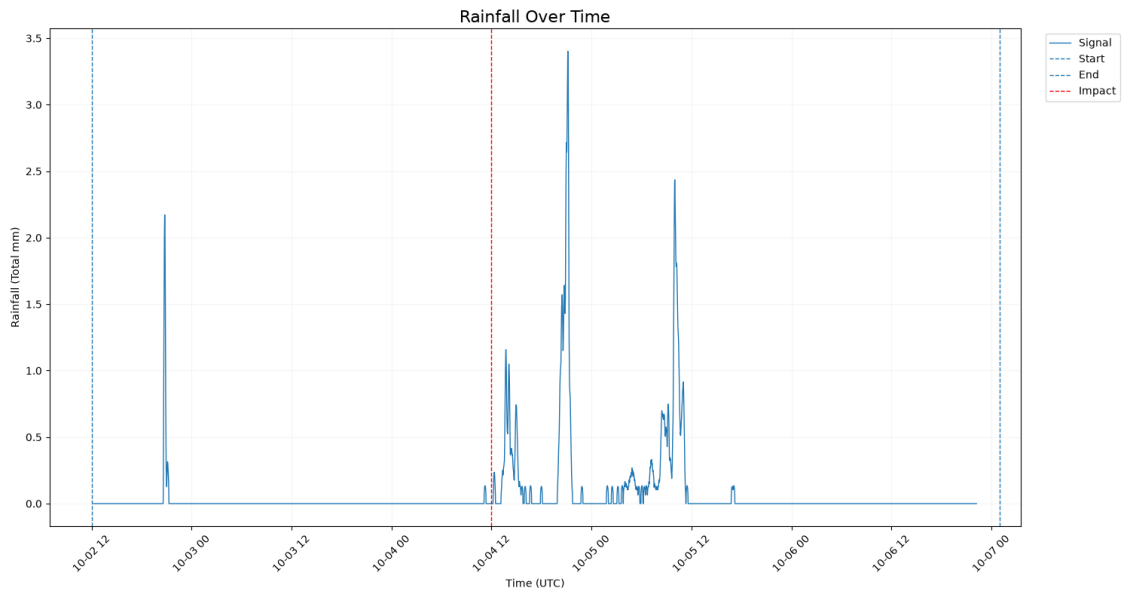
1.2 Executive Summary

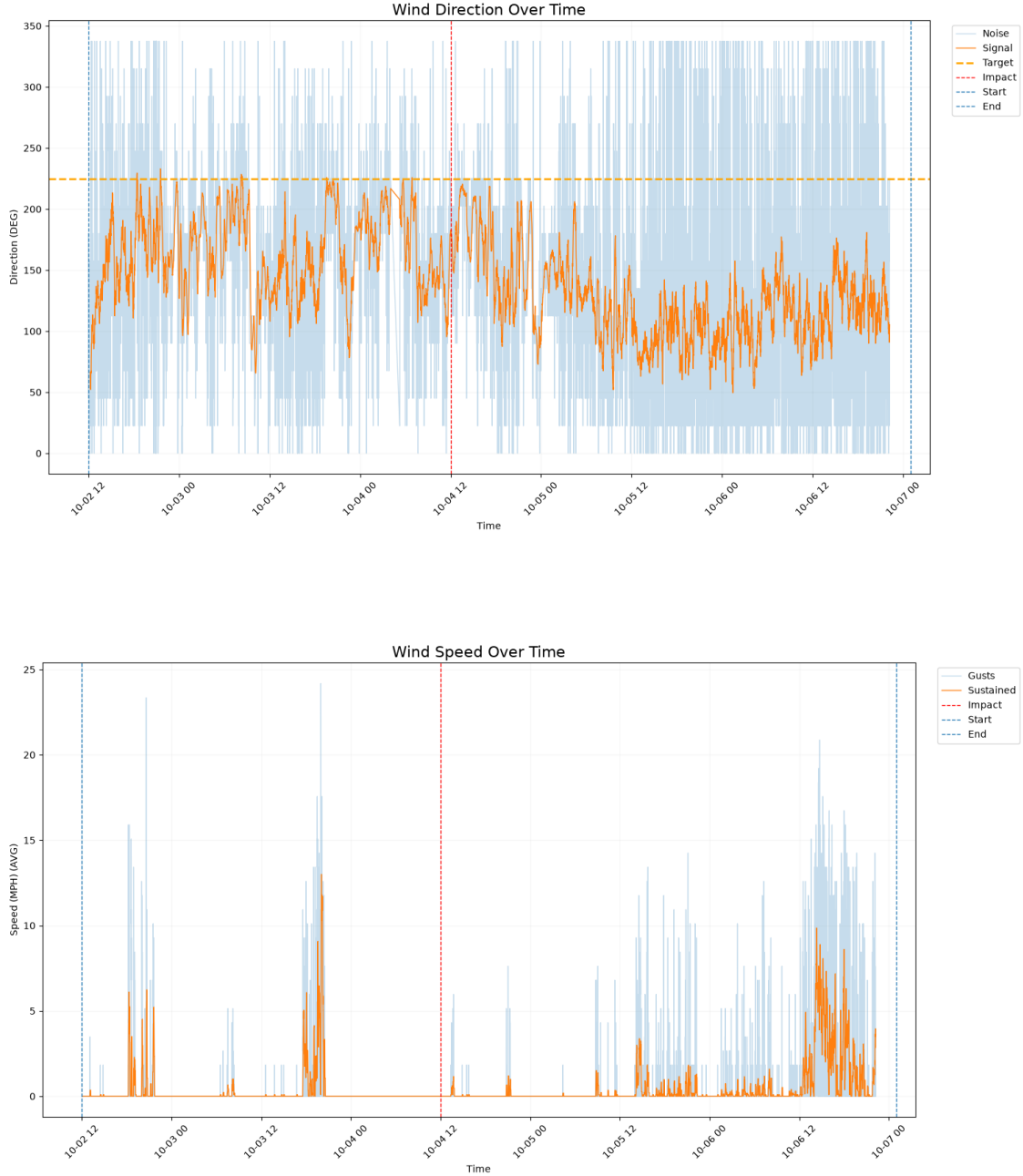
Low Pressure System 002 impacted the facilities envelope in a variety of ways. Primarily with excess external moisture via rain.

1.3 Meteorological Context

This system began impacting the air column surrounding the facility on 10-02-2026. On 10-3-2026, the first rainfall hit. What is generally interesting about this system is that the moisture and the rain started before the low pressure system footprint was in the overarching pattern of barometric pressure. (Meaning the first storms impacted the property during the primary ‘High’ driver before the low system officially pushes through the area.)



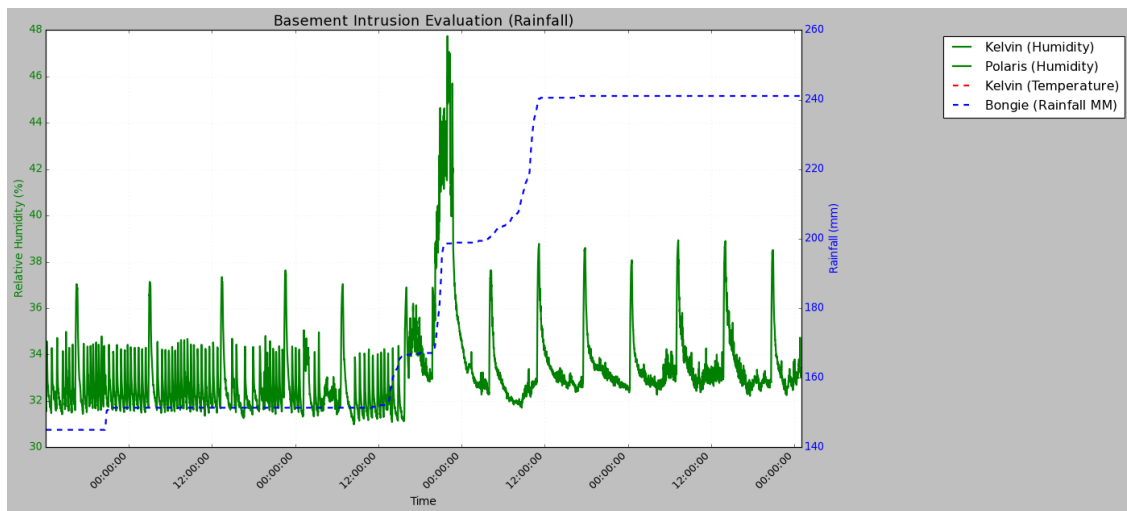
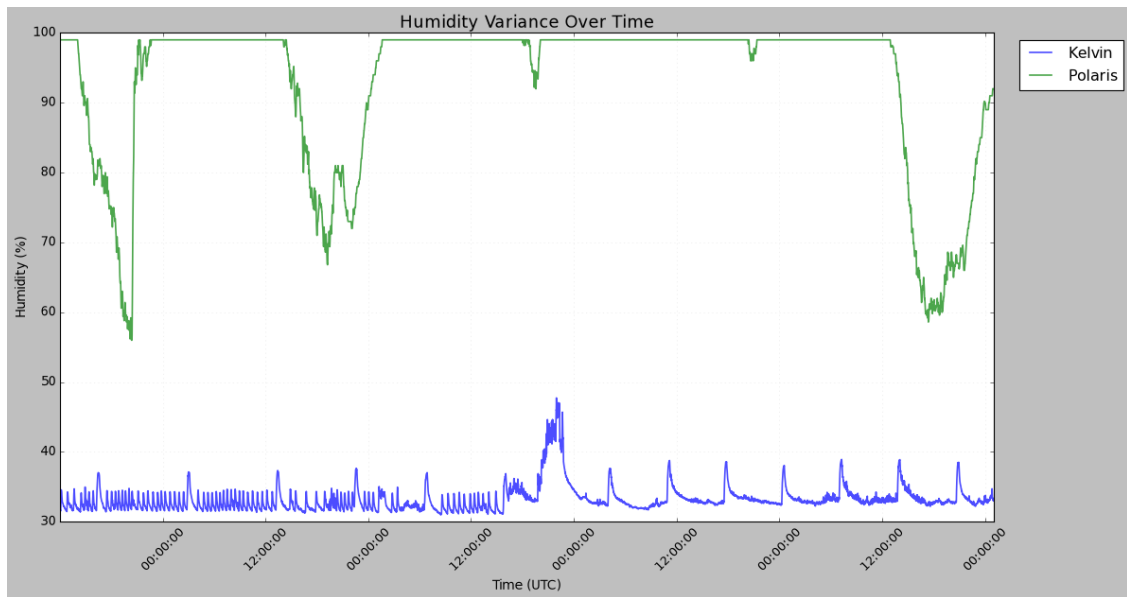


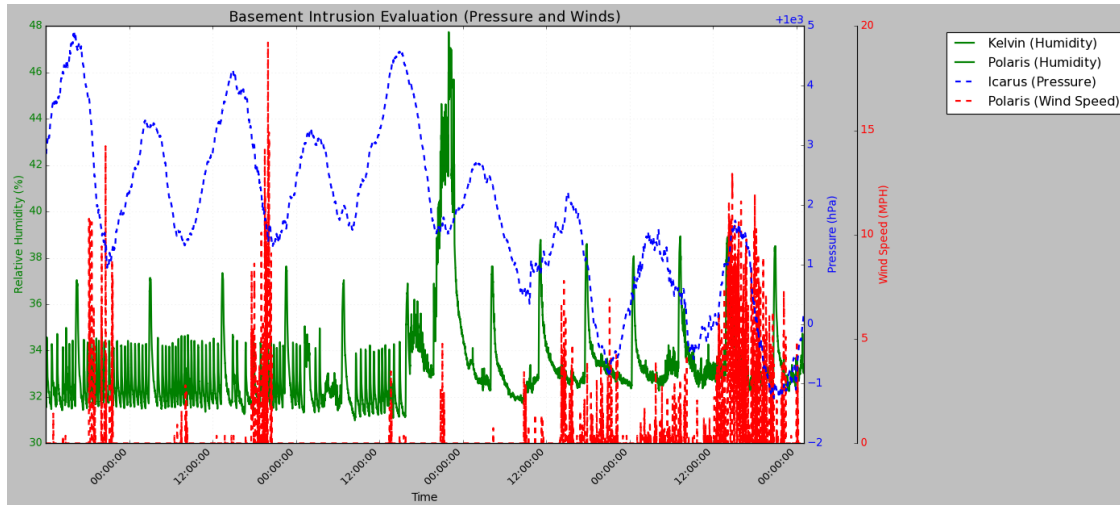


1.4 Basement Intrusion Post-Mortem

The basement space has been a primary concern since the project started; however this event highlighted the importance of studying that space. On 10/4/2026 around 16:00, water began impacting the air relative humidity faster than the dehumidifiers could extract the water.

Note: Manual initiative to dry up the puddles occurred around the west wall in the basement around 22:00:00 which reduced the amount of water evaporating into the basement air.





Note: Kelvin is stationed in the basement, Polaris on the rooftop, Icarus in the patio within the shade, and Bongie is the patio rain gauge station.

1.5 Overall Envelope Impact Assessment

- **Wind Exposure:** Maximum recorded sustained wind speed: 24 mph. (10-3-2026, Polaris)
 - Assessment: Minimal wind, no envelope damage.
- **Envelope Damage:** Significant water intrusion from the basement walls. Recommended investigation into potential solutions for preventing water intrusion through the west wall of the Basement.
- **Water Impacts:**
 - Very noticeable water intrusion in the basement through walls. Puddling near equipment and plastic totes. Potential causes include heightened water table near the west wall. Ground slopes toward and around the west wall and foundation of the envelope.
 - Significant humidity anomalies detected in the basement area (Elevated from 30% baseline to sustained nearly 50% during rainfall events and when water intrusion was at its peak. This was very similar to the last low pressure system **Tropical Storm Arthur**)

1.6 Equipment and Accuracy Assessment

The equipment overall handled the event very well with minor disturbances in the outside environment. Team was able to expand dashboard services, and monitor during several rainfall moments with no noticeable clock drift, lost packets (aside from station hardware migration to G2 in a few places but they were during calm hours.)

In contrast to LPS001 (Tropical Storm Arthur), this system was tracked with no clock drift, patio station remained online even though cloud cover extended a five day period (with noticeably shorter days), and the station range issues (limited by wifi but replaced by different radio in G2) were not present for the duration of this event.

1.7 A statement from the Architect

“This was the first time we got to test the second generation building science station design i
Charles Harding

1.7.1 Disclaimer

This report was generated based on data obtained by The Regulus Project in 2026. It is solely for scientific research purposes and is not an authoritative truth. The data within this report are the observations reported by the station mesh.

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